



**CATHOLIC JUNIOR COLLEGE**  
**JC2 PRELIMINARY EXAMINATIONS**  
**Higher 2**

**CANDIDATE  
NAME**

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**CLASS**

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**INDEX  
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## **BIOLOGY**

### **Paper 1 Multiple Choice**

**9744/01**

**16<sup>th</sup> SEPT 2022**

**1 hour**

Additional Materials: OMR Answer Sheet

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### **READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

**Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet** in the spaces provided unless this has been done for you.

There are **thirty multiple choice questions** on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate **OMR Answer Sheet**.

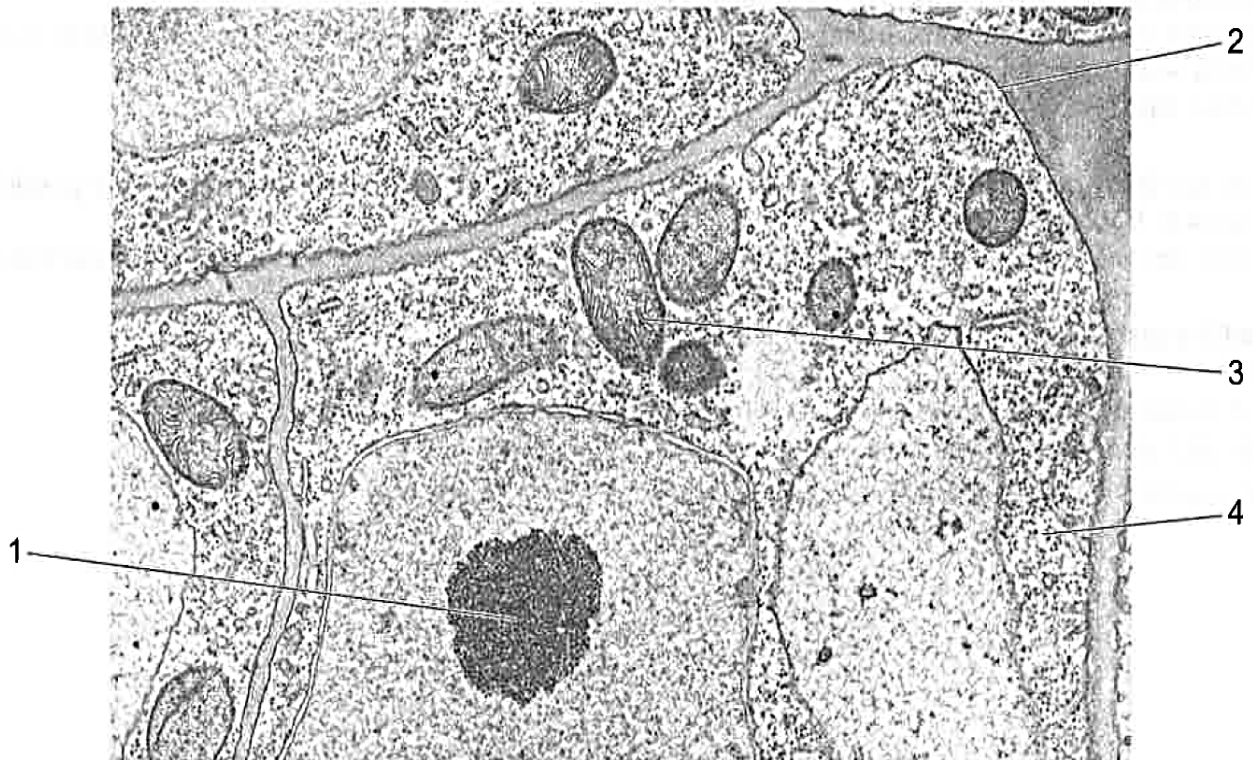
**Read the instructions on the OMR Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

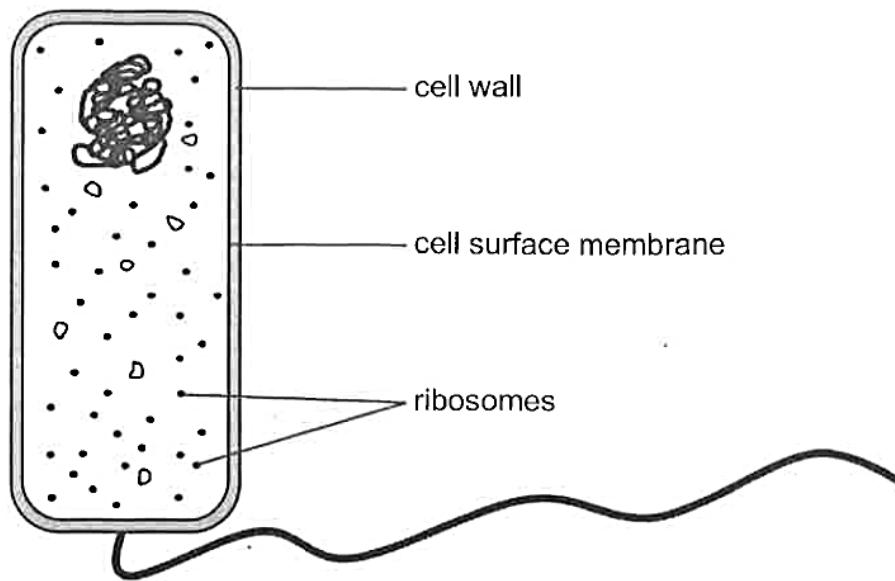
- 1 The electron micrograph shows parts of several eukaryotic cells.



Which row identifies a labelled location at which each process occurs?

|          | synthesis of ATP | synthesis of pyruvate | synthesis of rRNA |
|----------|------------------|-----------------------|-------------------|
| <b>A</b> | 1                | 3                     | 4                 |
| <b>B</b> | 4                | 4                     | 1                 |
| <b>C</b> | 3                | 4                     | 2                 |
| <b>D</b> | 2                | 3                     | 3                 |

2 The diagram shows a bacterial cell.



Which of the labelled structures contain chains of amino acids?

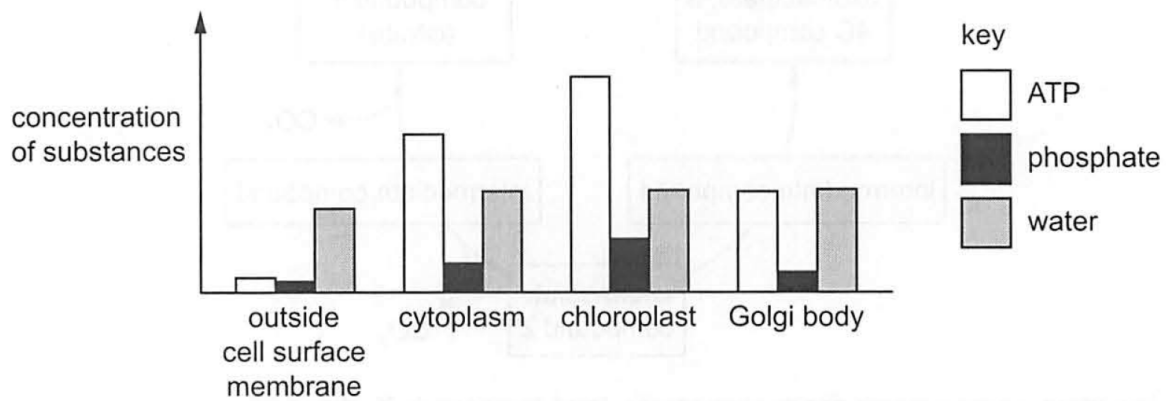
|          | cell wall | cell surface membrane | ribosome |
|----------|-----------|-----------------------|----------|
| <b>A</b> | √         | √                     | √        |
| <b>B</b> | √         | √                     | x        |
| <b>C</b> | √         | x                     | √        |
| <b>D</b> | x         | √                     | √        |

Key:

√ = contains chains of amino acids

x = does not contains chains of amino acids

- 3 The chart shows the concentration of some substances outside the cell surface membrane, in the cytoplasm, in the chloroplast and in the Golgi body of a plant cell.



Which statement about the direction of movement of these substances and the process by which they are moving is correct?

- A** ATP is leaving the chloroplast by facilitated diffusion, water is leaving the plant cell by osmosis
- B** Phosphate and ATP are entering the chloroplast and Golgi body by active transport
- C** Phosphate and ATP are leaving the Golgi body by facilitated diffusion, water is leaving the plant cell by osmosis
- D** Phosphate is entering the chloroplast by facilitated diffusion, water is entering the chloroplast by osmosis

- 4 The diagram below shows the synthesis of a new strand of DNA during interphase.

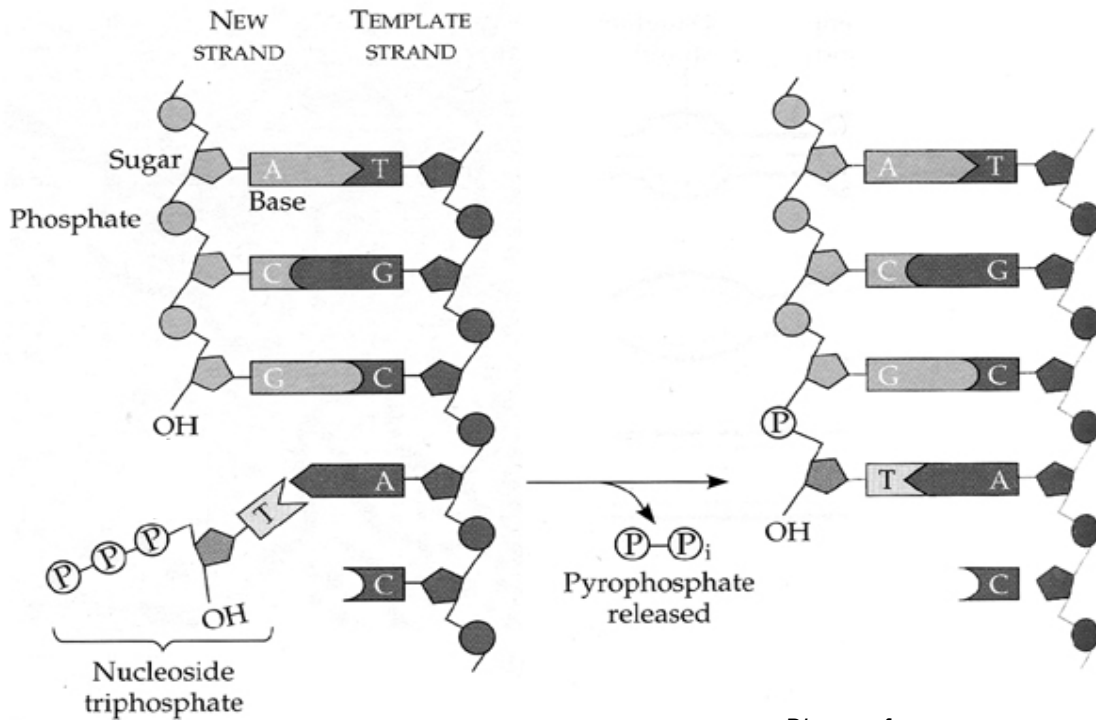


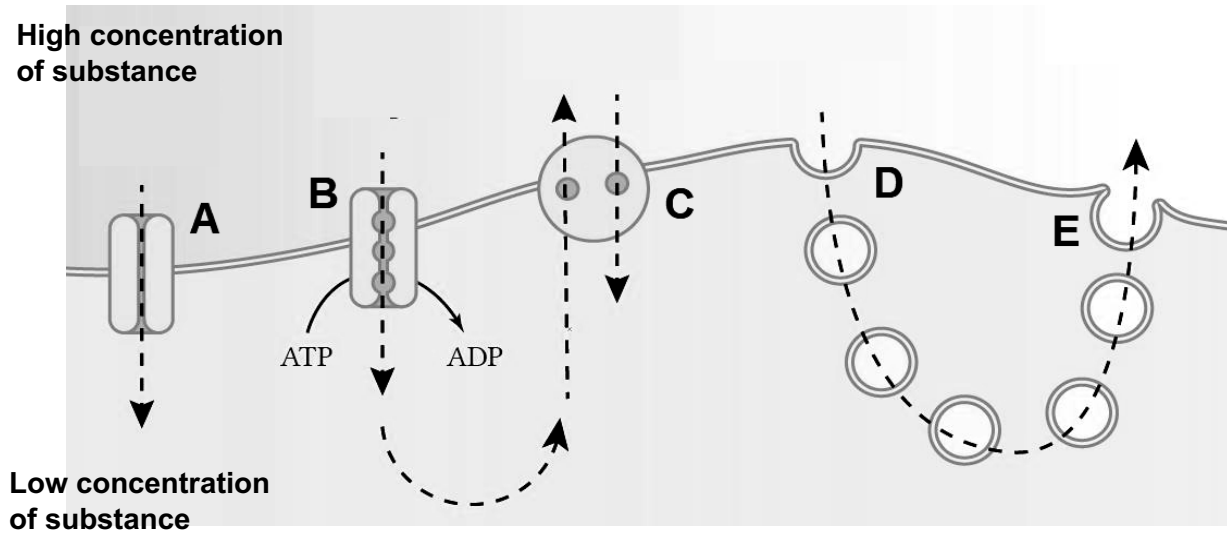
Diagram from  
<http://hshgp.genome.washington.edu>

With reference to the figure above, which of the following statements are true?

- I. New nucleotides are added at the 3' end of the new DNA strand via hydrolysis reactions.
- II. The resulting bonds that needed to be formed are phosphoester bonds and hydrogen bonds.
- III. Energy in the form of ATP is needed for the reaction to occur.
- IV. The bond formed between the sugar-phosphate backbone in the new DNA strand is called phosphodiester bond.
- V. Dephosphorylation of nucleoside triphosphate occurs as they are added onto the new DNA strand.

|          |            |
|----------|------------|
| <b>A</b> | I, II, IV  |
| <b>B</b> | I, III, V  |
| <b>C</b> | III, IV, V |
| <b>D</b> | II, IV, V  |

- 5 The figure below shows different mechanisms of transport across membrane, labelled **A** to **E**.



Which of the following shows a correct organisation of the mechanisms according to their features?

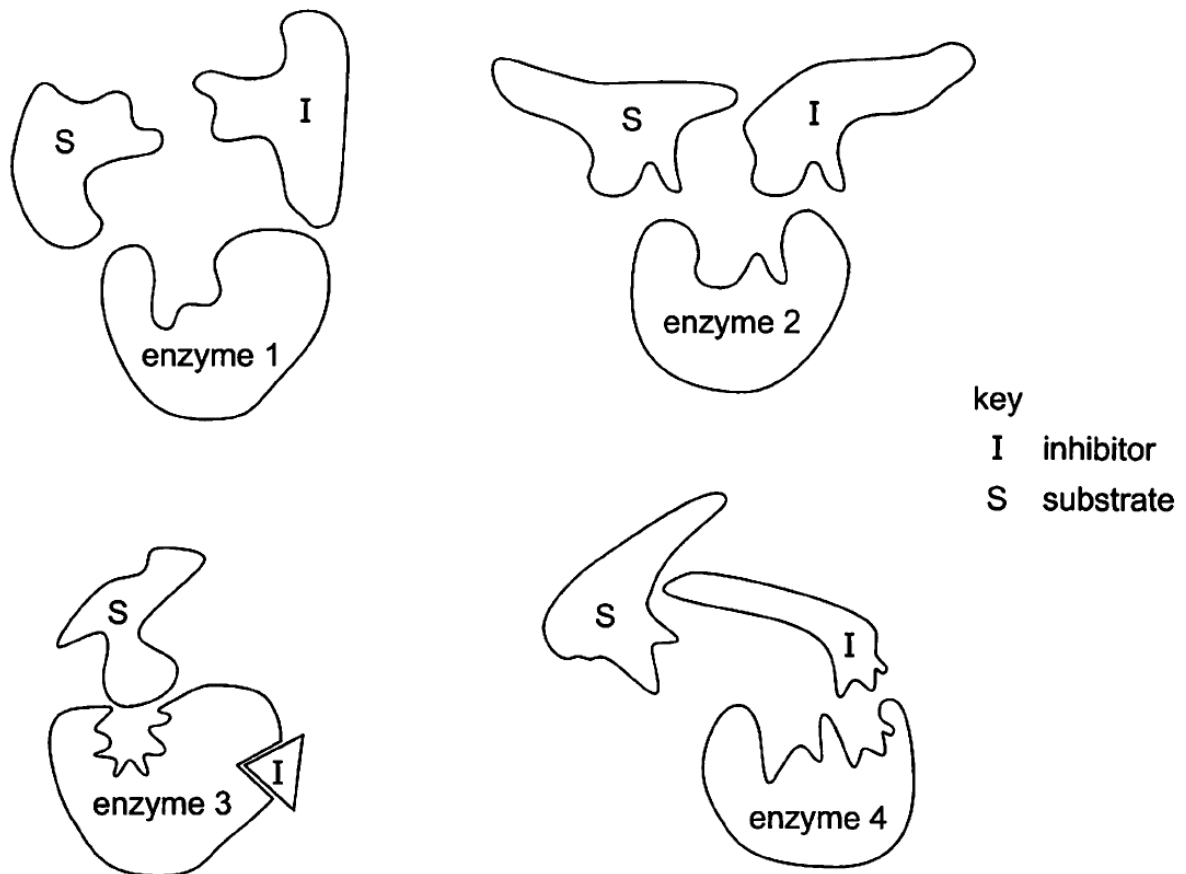
|          |                                                 |                                                    |
|----------|-------------------------------------------------|----------------------------------------------------|
| <b>A</b> | Transport molecules down concentration gradient | Transport molecules against concentration gradient |
|          | <b>A, C</b>                                     | <b>B, C, D, E</b>                                  |

|          |                                |                                    |
|----------|--------------------------------|------------------------------------|
| <b>B</b> | Transport polar molecules only | Transport non-polar molecules only |
|          | <b>A, B, C</b>                 | <b>D, E</b>                        |

|          |             |                |
|----------|-------------|----------------|
| <b>C</b> | Passive     | Active         |
|          | <b>A, C</b> | <b>B, D, E</b> |

|          |                           |                           |
|----------|---------------------------|---------------------------|
| <b>D</b> | Transport large molecules | Transport small molecules |
|          | <b>A, B, C</b>            | <b>D, E</b>               |

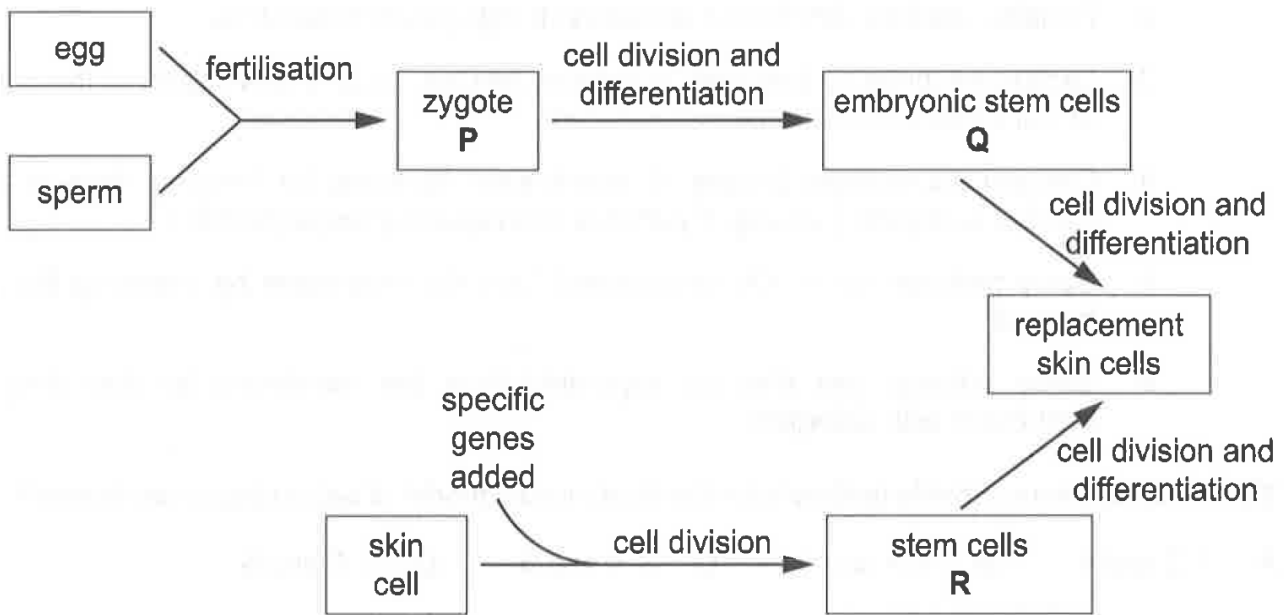
- 6 The diagram shows different types of reversible enzyme inhibition for four enzymes.



Which row correctly identifies the types of enzyme inhibition shown acting on enzymes 1, 2, 3 and 4?

|          | Competitive inhibition | Non-competitive inhibition |
|----------|------------------------|----------------------------|
| <b>A</b> | 1                      | 2, 3, 4                    |
| <b>B</b> | 1, 2, 4                | 3                          |
| <b>C</b> | 1, 2                   | 3, 4                       |
| <b>D</b> | 3, 4                   | 1, 2                       |

- 7 The diagram shows two ways in which replacement skin cells can be produced.



Which row correctly describes the stem cells **P**, **Q** and **R**?

|          | <b>P</b>    | <b>Q</b>            | <b>R</b>            |
|----------|-------------|---------------------|---------------------|
| <b>A</b> | pluripotent | multipotent         | totipotent          |
| <b>B</b> | pluripotent | induced pluripotent | induced pluripotent |
| <b>C</b> | totipotent  | pluripotent         | induced pluripotent |
| <b>D</b> | totipotent  | totipotent          | totipotent          |



- 8 A space probe brought back some living material from a distant planet. Proteins very similar to those found on Earth were present.

DNA of this living material, which exists as double helices, contains six different nucleic acid bases, identified as H, I, J, K, L and M. Among these, J, L and M were found to be pyrimidine bases.

Studies of the DNA from this living material gave the following results:

| ratio of bases in double-stranded DNA | numerical value |
|---------------------------------------|-----------------|
| H / J                                 | 1.00            |
| (H + I) / (J + M)                     | 1.02            |
| (H + K) / (J + M)                     | 1.14            |
| (H + K) / (J + L)                     | 1.01            |

Assuming that the base-pairing follows Chargaff's rule, what can be concluded from these data?

- (1) H base pairs with J
  - (2) H, I, K are likely to be purine bases
  - (3) It is probable that I base pairs with M, and K base pairs with L.
- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 9 The base analogue 5-bromouracil is a chemical mutagen that induces single-pair substitution in DNA called transitions (i.e. substitutions of one purine for another purine or one pyrimidine for another pyrimidine).

The genetic code (the code is given here in terms of the mRNA codons)  
second base

| first base | second base |     |     |     |     |     |     |     |     |   |  |   |
|------------|-------------|-----|-----|-----|-----|-----|-----|-----|-----|---|--|---|
|            | U           |     | C   |     | A   |     | G   |     |     |   |  |   |
|            | U           | UUU | Phe | UCU | Ser | UAU | Tyr | UGU | Cys | U |  | U |
|            |             | UUC |     | UCC |     | UAC |     | UGC |     | C |  | C |
|            |             | UUA |     | UCA |     | UAA |     | UGA |     | A |  | A |
|            |             | UUG |     | UCG |     | UAG |     | UGG |     | G |  | G |
|            | C           | CUU | Leu | CCU | Pro | CAU | His | CGU | Arg | U |  | U |
|            |             | CUC |     | CCC |     | CAC |     | CGC |     | C |  | C |
|            |             | CUA |     | CCA |     | CAA |     | CGA |     | A |  | A |
|            |             | CUG |     | CCG |     | CAG |     | CGG |     | G |  | G |
| A          | A           | AUU | Ile | ACU | Thr | AAU | Asn | AGU | Ser | U |  | U |
|            |             | AUC |     | ACC |     | AAC |     | AGC |     | C |  | C |
|            |             | AUA |     | ACA |     | AAA |     | AGA |     | A |  | A |
|            |             | AUG |     | ACG |     | AAG |     | AGG |     | G |  | G |
| G          | G           | GUU | Val | GCU | Ala | GAU | Asp | GGU | Gly | U |  | U |
|            |             | GUC |     | GCC |     | GAC |     | GGC |     | C |  | C |
|            |             | GUA |     | GCA |     | GAA |     | GGA |     | A |  | A |
|            |             | GUG |     | GCG |     | GAG |     | GGG |     | G |  | G |

Using the genetic code provided, which one of the following amino acid substitutions would you expect to be induced by 5-bromouracil with the highest frequency?

- A Met to Val
- B Met to Leu
- C Lys to Thr
- D Lys to Gln

- 10** Which of the following statements correctly accounts for HIV's ability to establish itself quickly in a human host?
- I. Interaction of Gp41 with CCR5 receptors trigger fusion and entry into the host cell
  - II. Reverse transcriptase allows for the synthesis of viral dsDNA that is integrated into the host genome.
  - III. Formation of an RNA template that is complementary to the genome of HIV allows for amplification in the synthesis of viral protein.
  - IV. Protease action converts the provirus into a mature virus
- A** I and II
- B** I and III
- C** II and III
- D** I, III and IV

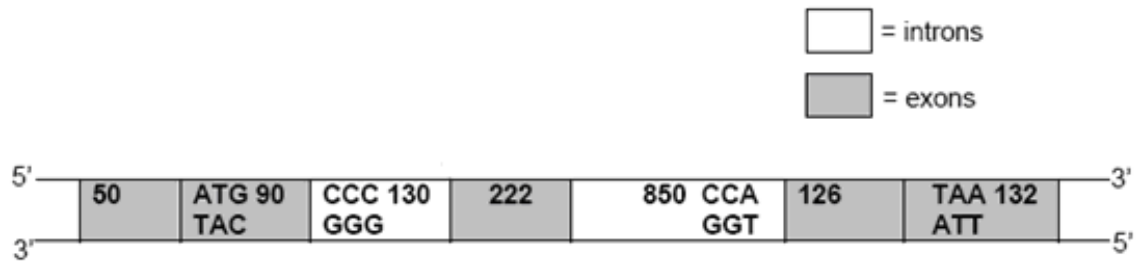
- 11** The table compares the structure and function of some elements of the *lac* operon.

|                      | made of<br>protein | binds to<br>protein | codes for<br>protein |
|----------------------|--------------------|---------------------|----------------------|
| <i>lac</i> repressor | 1                  | 2                   | 3                    |
| <i>lac</i> promoter  | 4                  | 5                   | 6                    |
| <i>lac</i> operator  | 7                  | 8                   | 9                    |
| <i>lacZ</i>          | 10                 | 11                  | 12                   |

Which combination of numbers links the four elements of the *lac* operon listed to their correct structures and functions?

- A** 1, 5, 6, 9 and 12.
- B** 1, 5, 8, 11 and 12.
- C** 2, 6, 7, 8 and 11.
- D** 3, 4, 8, 10 and 12.

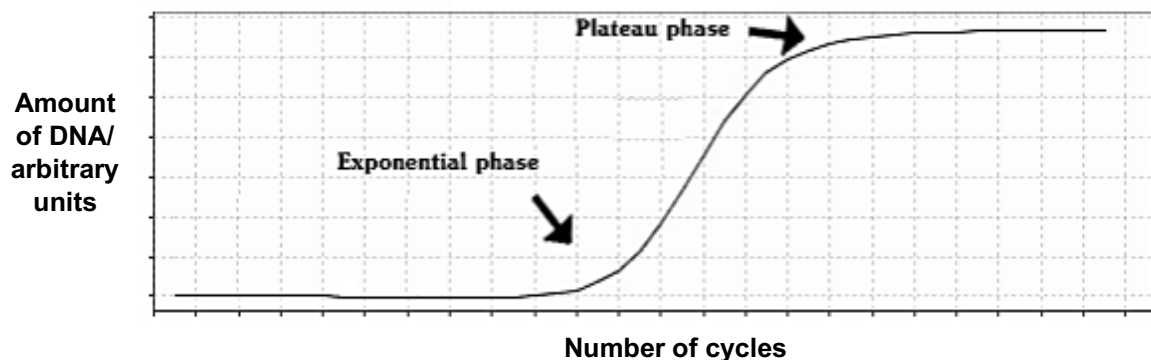
- 12 The diagram below shows the genomic structure of a human gene. The numbers within the boxes indicate the length of nucleotides of each region, inclusive of the bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are also indicated



What is the length (in nucleotides) of the primary RNA transcript (pre-mRNA) and how many amino acids are present in the protein?

|   | Length of primary RNA transcript | Number of amino acid in protein |
|---|----------------------------------|---------------------------------|
| A | 1 600                            | 146                             |
| B | 620                              | 146                             |
| C | 1 600                            | 206                             |
| D | 620                              | 206                             |

- 13 During the process of polymerase chain reaction (PCR), the amount of DNA synthesised can be traced using fluorescent probes and the measurements are shown in the following plot. The process initially goes through an exponential phase, followed by a plateau phase eventually.



Which of the following statements is **true**?

- A During the exponential phase, the number of DNA molecules synthesized after 15 cycles is  $15^2$ .
- B During the exponential phase, the temperature is always maintained at the optimum temperature of  $72^\circ\text{C}$  hence there is rapid amplification.
- C During the plateau phase, the reaction mixture is being depleted of ribonucleotides.
- D During the plateau phase, *Taq* polymerase may be denatured.

14 The diagram shows part of a small protein

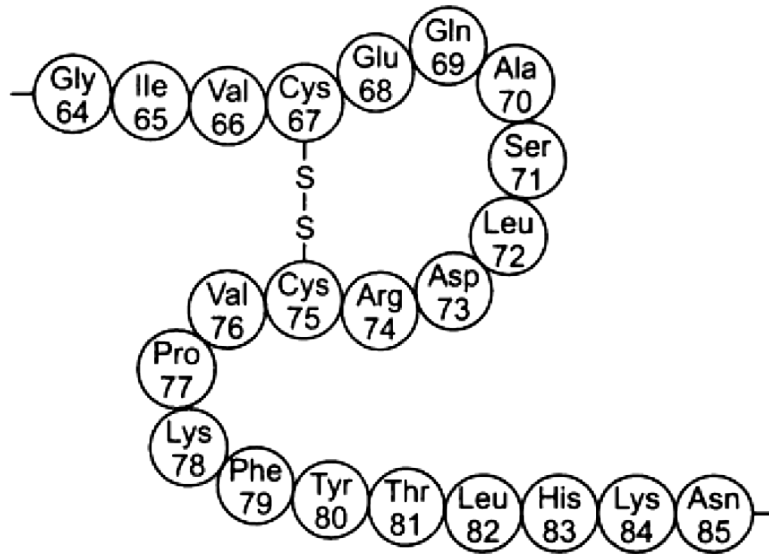


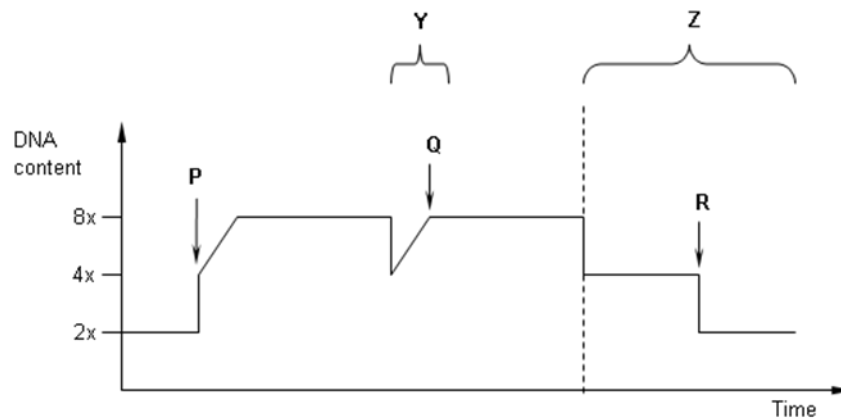
Fig. 14

Which mutations of the DNA coding for this part of the protein would always cause a change in shape of this part of the molecule?

- 1 a change in base sequence of the DNA triplet coding for the amino acid in position 74
- 2 a deletion from the DNA triplet coding for the amino acid in position 73
- 3 a substitution in the DNA triplet coding for the amino acid in position 69
- 4 an addition into the DNA triplet coding for the amino acid in position 70

- A** 1, 2, 3 and 4.
- B** 1, 3 and 4 only.
- C** 1 and 3 only.
- D** 2 and 4 only.

- 15 The graph represents the changes in the DNA content within a cell at different stages in the cell cycle.



Identify the events occurring at **P**, **Q** and **R**, and identify the stage where meiosis is occurring

|          | <b>P</b>      | <b>Q</b>      | <b>R</b>    | <b>Meiosis occurring at</b> |
|----------|---------------|---------------|-------------|-----------------------------|
| <b>A</b> | S phase       | Fertilisation | Cytokinesis | <b>Y</b>                    |
| <b>B</b> | Fertilisation | Interphase    | Cytokinesis | <b>Z</b>                    |
| <b>C</b> | S phase       | Prophase      | Telophase   | <b>Y</b>                    |
| <b>D</b> | Fertilisation | Metaphase     | Telophase   | <b>Z</b>                    |

- 16 Which of the following statements is incorrect regarding causative factors that are directly related to the cause of cancer.
- A** Chemical carcinogens mutate protooncogenes e.g. RAS which disrupts the regulation of the cell cycle and apoptosis.
  - B** ionizing radiation causes breaks in DNA that overwhelms DNA repair mechanisms.
  - C** HIV that lowers the immune system which may allow the infection of viruses e.g. Herpes virus that lead to cancer.
  - D** Hereditary loss of p53 gene leads to a decreased ability to suppress cancer.

- 17 In sweet pea plants, flower colour and pollen grain shape are controlled by two genes at different loci. Heterozygous plants with purple flowers and long pollen grains which were obtained from pure bred flowers were crossed to produce the offspring shown below.

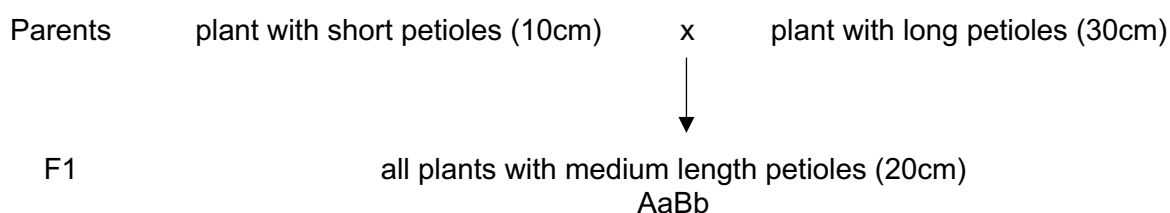
|                                     |     |
|-------------------------------------|-----|
| purple flowers, long pollen grains  | 15  |
| purple flowers, round pollen grains | 145 |
| red flowers, long pollen grains     | 53  |
| red flowers, round pollen grains    | 9   |

Which statement best explains the ratio of offspring observed?

- A** The gene loci for flower colour and pollen grain shape are located next to each other on the same chromosome.
- B** The alleles for purple flower colour are epistatic over the alleles for long pollen grain and the alleles for red flower colour are epistatic over the alleles for round pollen grain.
- C** The two gene loci are located on different sex chromosomes and crossing over occurred to give rise to recombinant gametes.
- D** The genes for red flower colour and long pollen grain are not assorting independently.
- 18 The length of the petiole (leaf stalk) in a type of flowering plant is controlled by two genes, A and B. These genes are at different loci on non-homologous chromosomes.

Plants with long petioles (30 cm) are homozygous dominant.  
Plants with short petioles (10 cm) are homozygous recessive.  
Each dominant allele contributes 5 cm to the petiole length.

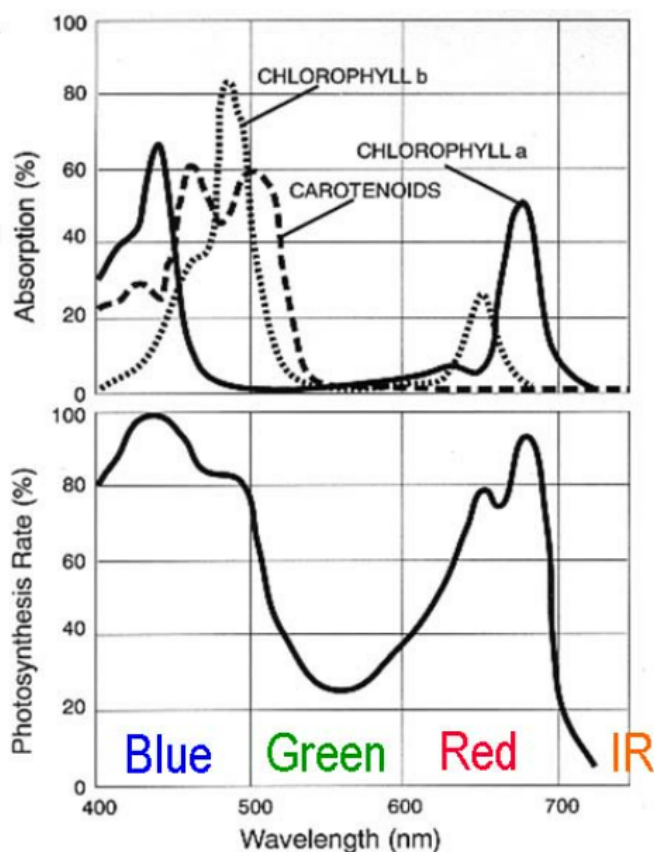
To obtain plants which have medium length petioles (20cm), a plant breeder carries out the cross below.



If the F1 generation plants with medium length petioles were allowed to cross, what proportion of their offspring would be expected to have medium length (20cm) petioles?

- A** 0.0625                      **B** 0.250                      **C** 0.375                      **D** 0.5

- 19 The two graphs below show the absorption and action spectrum for photosynthesis for the same plant.



Which statements correctly explain the relationship between the graphs above and the light dependent reactions?

- 1 The high absorption of blue light corresponds to the energy required to promote an electron in photosystem 2 of non-cyclic photophosphorylation.
- 2 The poor absorption of green light by all three pigments provides only enough energy for cyclic photophosphorylation to occur.
- 3 Carotenoids increases the overall rate of photosynthesis by extending the range of wavelengths absorbed at the blue-green range.
- 4 There is more energy in blue and red light which is why there are higher absorption rates of these wavelengths of light for photosynthesis.

**A** 3 only

**B** 2 and 3 only

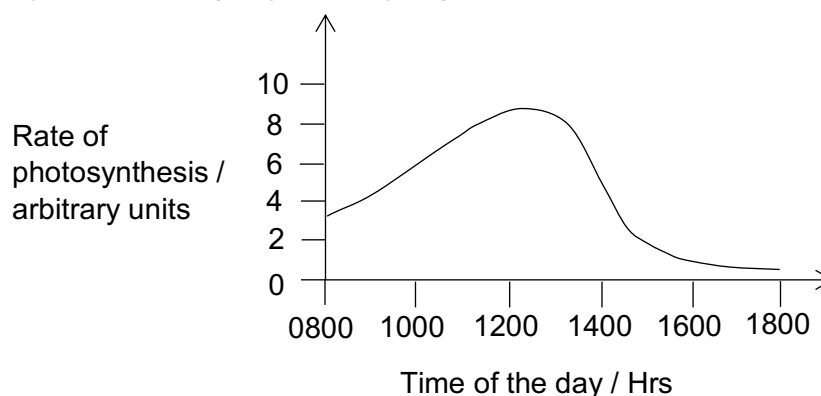
**C** 1, 2 and 4

**D** 1, 3 and 4



- 20** Crassulacean acid metabolism (CAM) is a mechanism evolved in certain plants, such as the pineapple, that are adapted to arid conditions. In these plants, stomata are closed during the day to prevent water loss and opened at night to take in carbon dioxide, which is fixed and stored as an organic acid. The organic acids then function as sources of carbon dioxide for photosynthesis to take place during the day.

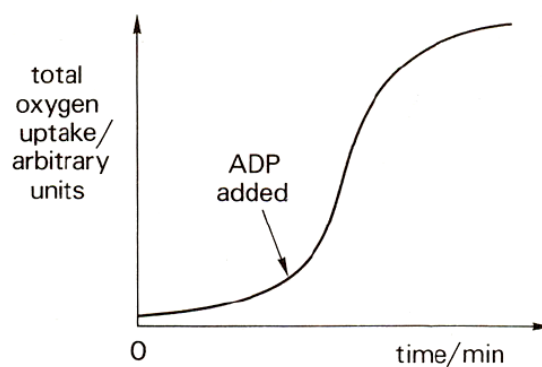
The rate of photosynthesis during a typical day is given below.



Which of the following is most likely to be the limiting factor for photosynthesis in plants utilising CAM at 1000 Hrs and 1400 Hrs?

|          | 1000 Hrs       | 1400 Hrs       |
|----------|----------------|----------------|
| <b>A</b> | Temperature    | Water          |
| <b>B</b> | Light          | Carbon dioxide |
| <b>C</b> | Water          | Temperature    |
| <b>D</b> | Carbon dioxide | Light          |

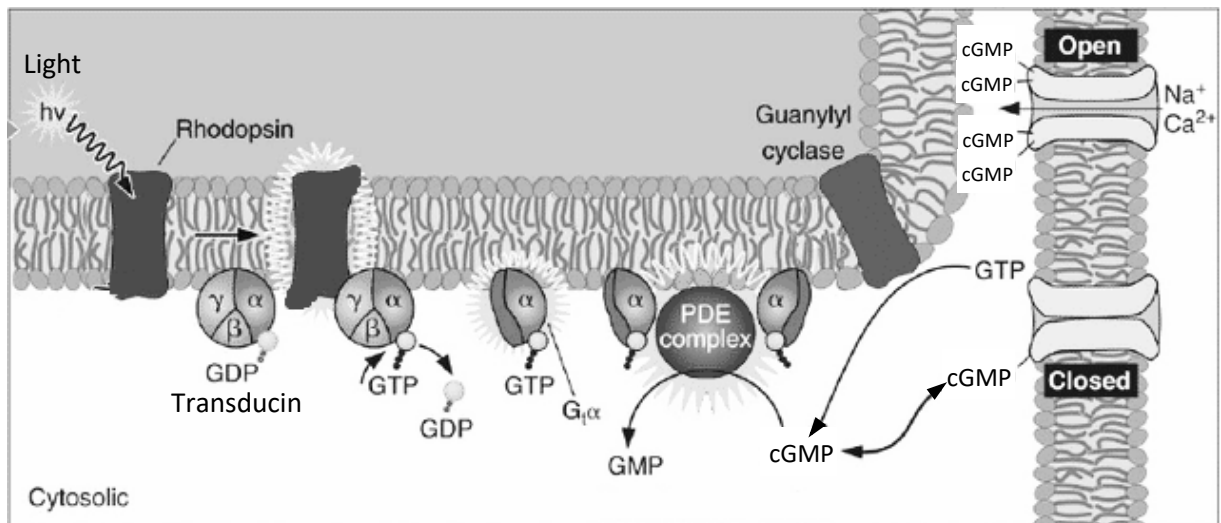
- 21** The graph shows the total oxygen uptake by a sample of homogenised liver tissue over a period of time.



Which statement explains the sharp rise in oxygen uptake after the addition of ADP?

- A** Electron transport is coupled to the phosphorylation of ADP.
- B** More protons move directly across the phospholipid bilayer of the cristae, generating the proton pool in the intermembrane space.
- C** Slower utilisation of reduced NAD and FAD causes molecular oxygen to be reduced to water.
- D** Oxidative phosphorylation occurs as protons are pumped back into the matrix through ATP synthase, phosphorylating ADP.
- 22** Which statement explains why the energy yield from anaerobic respiration in yeast is lower than that from aerobic respiration?
- A** Decarboxylation of pyruvate to form ethanol loses energy to the environment.
- B** ATP produced via conversion of pyruvate to ethanol is much fewer compared to oxidative phosphorylation.
- C** Products of anaerobic respiration are not fully oxidised and retain a lot of chemical energy.
- D** Energy is required in order to produce  $\text{NAD}^+$  required for ATP production.

- 23** Rhodopsin is a type of G-protein linked receptor which is activated by light instead of ligand, found in the rod cells of the retina in the eye. The figure below shows the cell signalling pathway involving rhodopsin.



Which of the following statements correctly describe the cell signalling pathway of rhodopsin?

- 1 Light changes the conformation of rhodopsin.
- 2 Transducin is a G-protein.
- 3 Activation of rhodopsin leads to an increased level of cGMP levels inside the cell.
- 4 Guanyl cyclase is involved in the termination of the cell signalling pathway.
- 5 The reception of light leads to opening of the cGMP-gated ion channels, leading to influx of Na<sup>+</sup> and Ca<sup>2+</sup> ions into the cell.

- A** 1 and 3  
**B** 1, 2 and 4  
**C** 1, 2 and 5  
**D** 2, 3 and 5

- 24** Different species of bacteria are often referred to as different “strains”. Which of the following statements best explains why a bacteria strain is the smallest unit that can evolve?

- A** It is the smallest unit for which the gene pool of a population of bacteria is complete, where all alleles can be found.  
**B** It is the smallest unit for which conjugation between one strain and other strains in the same habitat can result in a selective advantage and therefore change in allelic frequencies.  
**C** It is the smallest unit for which random mutations can present opportunities for new strains to be formed from the original population.  
**D** It is the smallest unit for which there can be changes in allelic frequencies for a population of bacteria to adapt to a change in selection pressure.

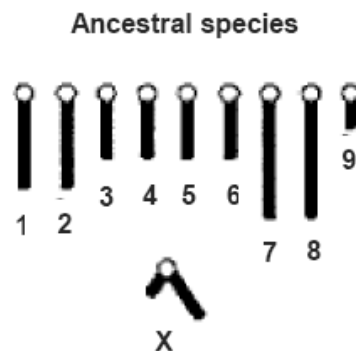
- 25** Vitamin C is an important molecule required for the biosynthesis of collagen and act as antioxidants to radicals in the body. Unlike many animals, humans have lost the ability to synthesise vitamin C endogenously, such that it is now considered an essential dietary component.

Scientists have proposed some hypotheses about why some mammals and primates, including human ancestors, have lost the ability to synthesise vitamin C.

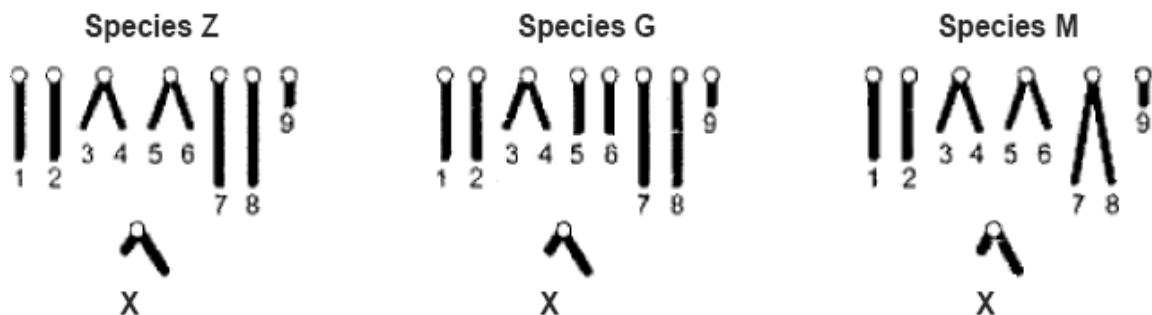
Which of the following statements does **not** support the loss of ability to synthesise vitamin C was due to natural selection?

- A** In the habitat of the mammal ancestors, there were abundant fruits containing vitamin C. As the ability to synthesise vitamin C costs resources and energy, it became selectively disadvantageous instead.
- B** Various mutations occurred in the genes responsible for vitamin C synthesis over a period of time which persisted due to chance. When vitamin C became rare, only those which were able to find food sources of vitamin C survived and reproduced more successfully.
- C** The mutation of the gene involved in vitamin C synthesis was linked to the expression of an alternative protein transporter allowing for storing an alternative form of vitamin C, which helped resist fluctuations in nutrient intake.
- D** A decreased level of vitamin C led to generation and accumulation of more radicals attacking the DNA. This might have led to a higher rate of molecular evolution due to DNA oxidation, consequently enabling faster adaptation to environmental changes.

- 26** In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, or rearrangement of the genetic material within chromosomes. One ancestral species has the following haploid set of chromosomes.



As the changes in chromosomes accumulate, three different species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



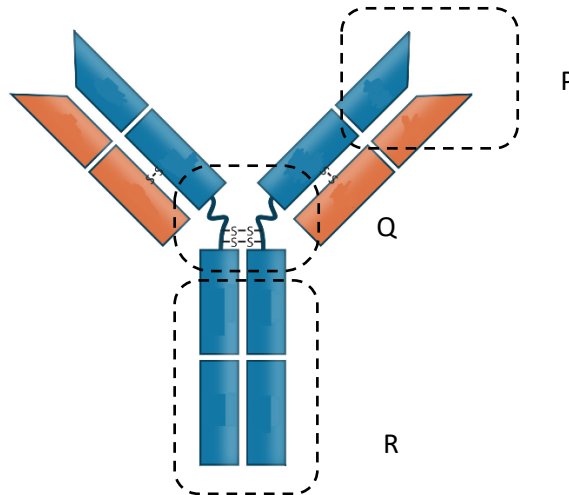
The most likely order of evolution of these species is

- A** ancestral species, species Z, species G, species M.
- B** ancestral species, species G, species M, species Z.
- C** ancestral species, species M, species G, species Z.
- D** ancestral species, species G, species Z, species M.

- 27 Which of the following correctly describes the relationship between the functions of B cells and CD4 T cells?

|          | B cells                                                                                                                                  | CD4 T cells                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | binds to antigen presenting cells and secrete cytokines to activate naïve CD4 T cells to divide and differentiate into cytotoxic T cells | secrete granzymes and perforin to induce apoptosis in macrophages which ingested bacteria, causing destruction of pathogens |
| <b>B</b> | binds to antigen presenting cells and secrete cytokines to activate naïve CD4 T to divide and differentiate into T helper cells          | secrete other cytokines which activates other immune cells, including the B cell and other T cells                          |
| <b>C</b> | able to act as antigen presenting cells to activate naïve CD4 T cells to divide and differentiate into T helper cells                    | secrete chemokines to recruit macrophages to accumulate at the site of infection                                            |
| <b>D</b> | able to act as antigen presenting cells to activate naïve CD4 T cells to divide and differentiate into cytotoxic T cells                 | secrete chemokines to recruit macrophages to accumulate at the site of infection                                            |

28 The diagram shows an antibody molecule.



Which of the following statements are true about the production of antibody diversity?

- 1 Presence of multiple variable gene segments in the gene encoding for region P allow for combinatorial diversity.
- 2 Variation in the structure of Q where different sequences are translated as a result of alternative splicing.
- 3 Class switching in region R, with the genes coding for region P remaining constant.
- 4 There can be two different types of heavy chains in a single antibody that originated from the same clone of plasma cell.
- 5 Affinity maturation through the process of somatic hypermutation in the production of mature B and T cells.

**A** 1 and 3

**B** 1, 3 and 5

**C** 2, 4 and 5

**D** 1, 4 and 5

- 29** Increased fossil fuel usage is one of the human activities contributing the most to atmospheric carbon dioxide in causing climate change. One of the ways scientists track this is by monitoring the isotopic fingerprint of the carbon present in today's atmosphere. Different kinds of carbon-containing material have different relative amounts of "light" carbon-12, "heavy" carbon-13, and radioactive carbon-14 which decays over time.

Which statement would support the conclusion that the increase in carbon dioxide emissions found in today's atmosphere is mostly due to human activities?

- A** The amount of carbon-12 in the atmosphere would increase since burning of fossil fuel releases high volume of trapped carbon.
  - B** The percentage of carbon-12 and carbon-13 in the atmosphere would increase as radioactive carbon-14 decays into these isotopes.
  - C** The percentage of carbon-14 in the atmosphere would increase as carbon dioxide released by fossil fuel usage have a higher percentage of radioactive carbon-14.
  - D** The percentage of carbon-14 in the atmosphere would decrease as fossil fuels contain relatively less amount of radioactive carbon-14.
- 30** Climate change is projected to cause a drastic loss in biodiversity worldwide, especially in more sensitive ecosystems such as the forests, tundras and alpine areas.

Which of the following statements correctly explains why these areas are at higher threat due to climate change?

- 1 Tropical rainforests are highly dependent on sufficient rainfall throughout the year.
  - 2 Tundra and alpine areas have been highly deforested by human activities.
  - 3 Organisms living in forests live in a narrow temperature range.
  - 4 The shift of organisms towards the poles has high probability of upsetting the ecology of the tundra and alpine areas.
- A** 1 and 2
  - B** 1 and 4
  - C** 2, 3 and 4
  - D** 1, 2, 3 and 4

**[End of Paper 1]**